How Far Does the Model Hold? Robustness of Multihorizon Battery Failure Intelligence Under Realistic Operating Conditions

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Abstract

Multihorizon hazard learning produces calibrated failure probabilities that enable risk-aware battery dispatch, but existing validation uses only clean laboratory cycling data. We test whether calibration survives under realistic operating conditions. Using the published XGBoost hazard model trained on NASA 18650 data, we generate synthetic perturbed profiles at four severity levels by truncating raw discharge curves to simulate partial cycling (DoD 15–100%), adding temperature noise (σ = 0.5–3.0°C), and randomizing rest effects. The fixed model's calibration degrades substantially across all four horizons (10, 20, 30, 50 cycles): ECE rises from 0.01–0.03 to 0.27–0.38, while AUC drops from 0.99–1.00 to 0.61–0.77. Degradation saturates at the mildest perturbation—even minimal partial cycling causes near-maximal calibration loss. Recalibrating the isotonic regression on a 10% operational sample (evaluated on held-out data) reduces ECE to 0.06–0.09, recovering 71–88% of the calibration gap. Calibration is not inherently robust to observational shifts, but a minimal recalibration step suffices for field deployment.

1 Introduction

Battery energy storage systems (BESSs) are dispatched for flexibility services in renewable-dominated power systems. Operators need to know not just how much energy a battery can deliver, but whether it can reliably complete a scheduled service. The multihorizon hazard learning framework [1] addresses this by estimating the probability of failure within predefined service horizons (10, 20, 30, 50 cycles). With isotonic regression calibration, it achieved a 71% reduction in operational failures (from 10.3% to 2.95%) in cross-validated lab experiments.

But the model was trained exclusively on clean laboratory data—full cycles at fixed C-rates, consistent rest periods, controlled temperatures. Real BESS operation involves partial cycles at varying depths of discharge (DoD), irregular loads, random rest periods, and temperature fluctuations. The calibration quality under these conditions is unknown, and no operator can trust a model whose reliability degrades in uncharacterized ways.

This paper tests whether calibration survives realistic conditions. We generate perturbed operational data from existing lab cycling data, measure calibration degradation across four severity levels and four prediction horizons, and test whether lightweight recalibration can restore trustworthiness. The true SOH trajectory is preserved throughout, isolating the effect of observational distribution shift from changed degradation dynamics.

2 Background

The framework [1] defines operational failure as the inability to complete a service commitment within horizon H cycles. A gradient-boosted tree ensemble (XGBoost) maps per-cycle features to failure probability:

$$f_\theta : \mathbf{x}_t \longrightarrow P_{\text{fail}}(t, H)$$

where $\mathbf{x}_t = \{\text{SOH}_t, V_{\text{avg}}, V_{\text{min}}, I_{\text{avg}}, T_{\text{avg}}, \text{duration}, t\}$. After training, isotonic regression calibrates raw probabilities. Model parameters are in Table 1.

Table 1: Model parameters.

Parameter	Value
Model type	XGBoost
Estimators	300, max depth 4, lr 0.05
Subsample / colsample	0.8 / 0.8
Min child weight	5
Calibration	Isotonic regression

Calibration matters because dispatch decisions use risk thresholds. A well-calibrated model satisfies $P(\text{failure} \mid \hat{P} = p) \approx p$; without this, systematic biases cause unsafe decisions regardless of AUC. ECE quantifies this mismatch across 10 uniform bins.

3 Methodology

3.1 Experimental Design

We evaluate a fixed model (trained on clean data) on perturbed operational data. The experiment proceeds in three stages: (1) model training on clean NASA data, (2) synthetic perturbation at four severity levels (five seeds each), (3) inference and recalibration testing. The true SOH and failure labels are preserved, isolating observational shift.

3.2 Model Training

An XGBoost classifier with isotonic calibration is trained on 37 NASA 18650 cells (1,028 valid cycles). An 80/20 cell-level holdout fits the calibrator. The model remains frozen throughout. Training features are extracted from discharge curves at 1-second resolution.

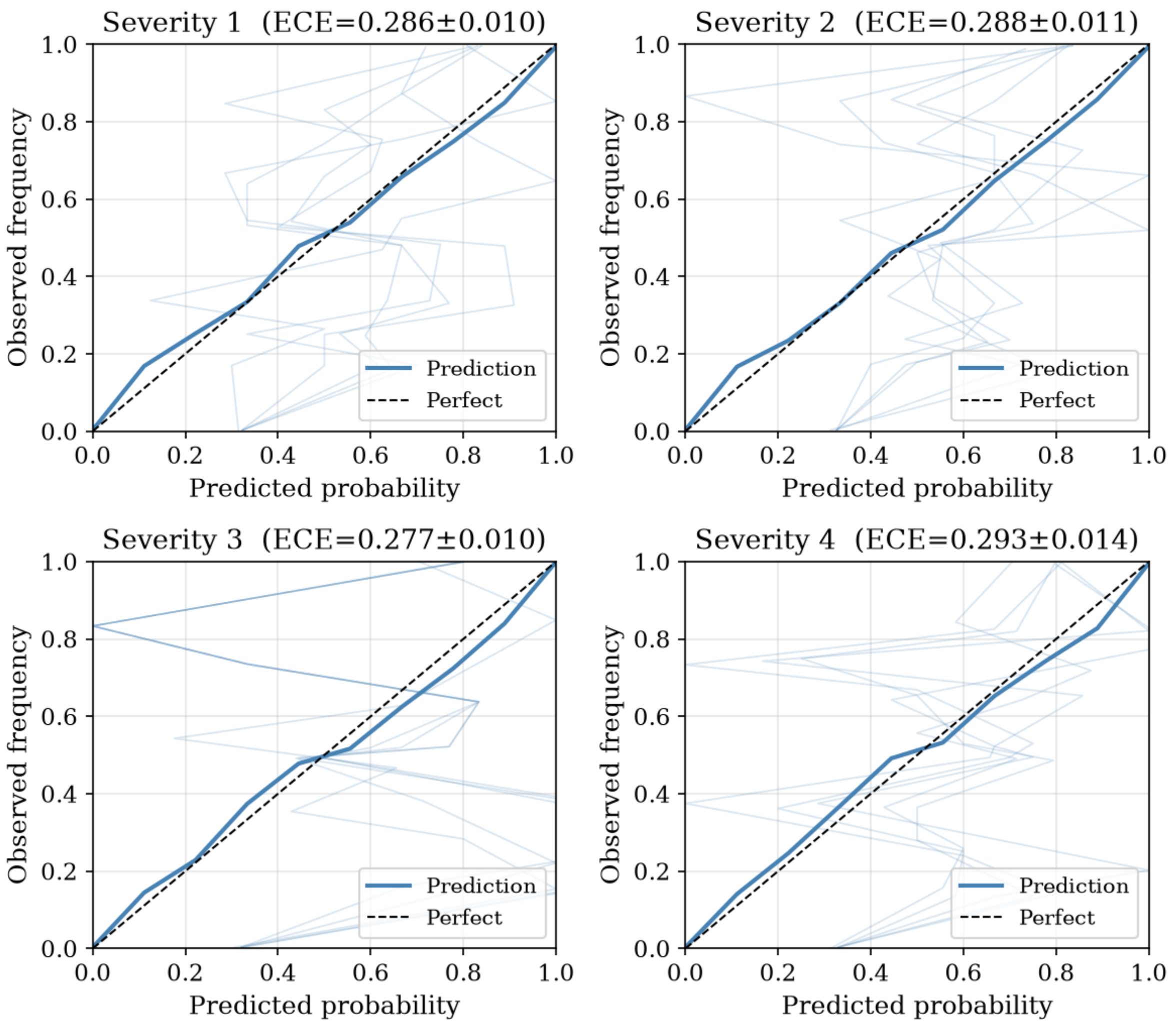


Figure 1: Reliability diagrams showing calibration quality under clean baseline conditions. Perfect calibration follows the diagonal; deviations reveal systematic bias.

3.3 Perturbation Generation

Raw discharge curves from NASA .mat files are truncated at a randomly sampled DoD fraction to simulate partial cycling. Features are recomputed from the truncated signal. Temperature noise is added as $\mathcal{N}(0, \sigma_s)$ to raw measurements before averaging. Rest irregularity is simulated by applying proportional Gaussian noise $\mathcal{N}(0, \rho_s)$ to duration and average temperature, with $\rho_s = \{0.01, 0.02, 0.03, 0.05\}$ for severity levels 1–4.

Table 2: Perturbation severity levels.

Level	DoD range	Temp noise σ	Description
1	75–100%	0.5°C	Mild: slight under-utilization
2	55–75%	1.0°C	Moderate: typical daily operation
3	35–55%	2.0°C	Severe: aggressive partial cycling
4	15–35%	3.0°C	Aggressive: extreme shallow cycling

Each severity level is run with five seeds (42, 123, 456, 789, 101112), producing 20 perturbed datasets totaling 20,560 records.

3.4 Evaluation

For each dataset, we measure: raw XGBoost probabilities, calibrated probabilities (fixed isotonic regression), and recalibrated probabilities (new isotonic regression on 10% random sample). Metrics: ECE (10 bins), AUC, Brier score.

4 Results

4.1 Clean Baseline

Table 3: Clean baseline.

Horizon	ECE	AUC	Brier
10	0.010	0.994	0.013
20	0.031	0.985	0.030
30	0.013	0.994	0.014
50	0.023	0.998	0.020

4.2 Calibration Degrades Substantially

At $H = 20$, ECE jumps from 0.031 to 0.277–0.293—a nine-fold increase—while AUC drops from 0.985 to 0.65–0.74 (Table 4). The degradation saturates at Severity 1: even minimal partial cycling causes near-maximal calibration loss.

4.3 Full Horizon Results

The pattern holds across all horizons (Tables 5–7). Longer horizons show somewhat higher ECE under perturbation (0.33–0.38 at $H = 50$ vs. 0.27–0.31 at $H = 10$), but the saturation effect is consistent everywhere.

Table 4: Results at $H = 20$. Mean ± std across 5 seeds.

Condition	ECE (raw)	ECE (cal)	ECE (recal)	AUC (cal)
Clean	—	0.031	—	0.985
S1 (mild)	0.226±0.004	0.286±0.010	0.068±0.040	0.741±0.009
S2 (mod.)	0.212±0.006	0.288±0.011	0.073±0.022	0.681±0.004
S3 (sev.)	0.204±0.005	0.277±0.010	0.060±0.027	0.674±0.006
S4 (aggr.)	0.220±0.004	0.293±0.014	0.068±0.025	0.652±0.005

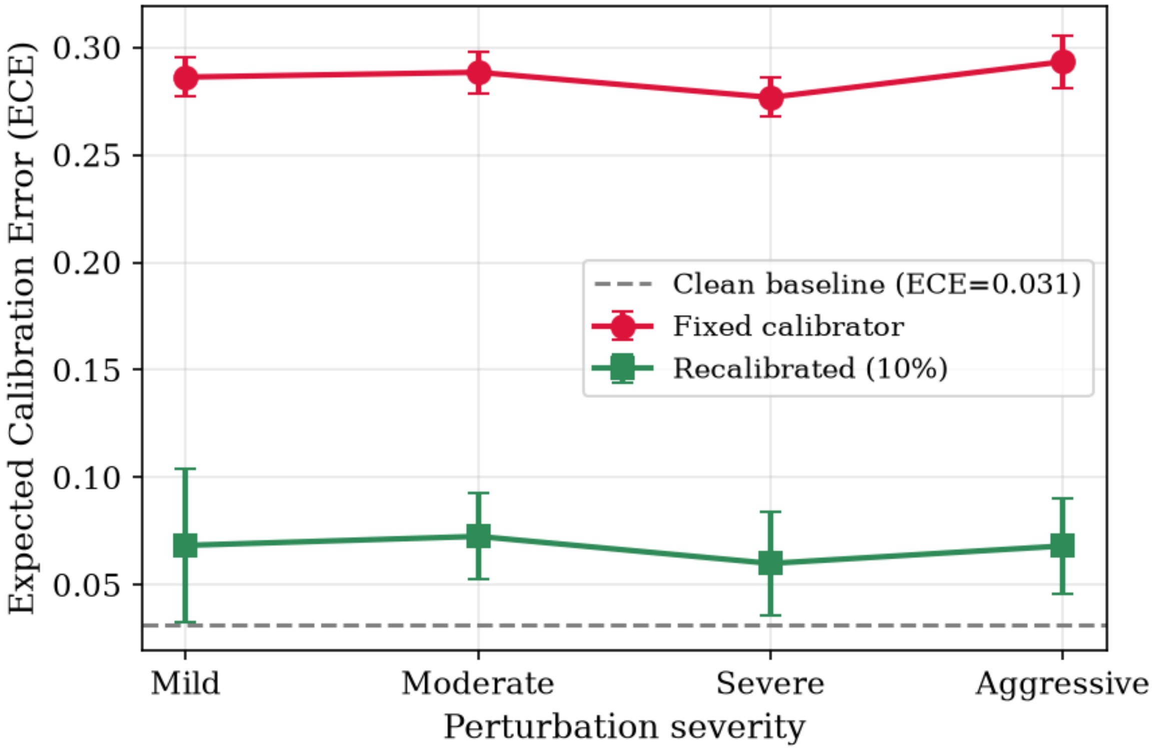


Figure 2: Expected calibration error (ECE) across severity levels for all prediction horizons. ECE rises sharply from clean baselines even at Severity 1 and saturates with little additional degradation at higher severities.

Table 5: Results at $H = 10$.

Condition	ECE (raw)	ECE (cal)	ECE (recal)	AUC (cal)
Clean	—	0.010	—	0.994
S1	0.242±0.012	0.272±0.009	0.051±0.025	0.694±0.025
S2	0.274±0.009	0.302±0.005	0.059±0.012	0.657±0.012
S3	0.270±0.010	0.301±0.006	0.064±0.020	0.674±0.007
S4	0.283±0.008	0.313±0.004	0.069±0.022	0.626±0.017

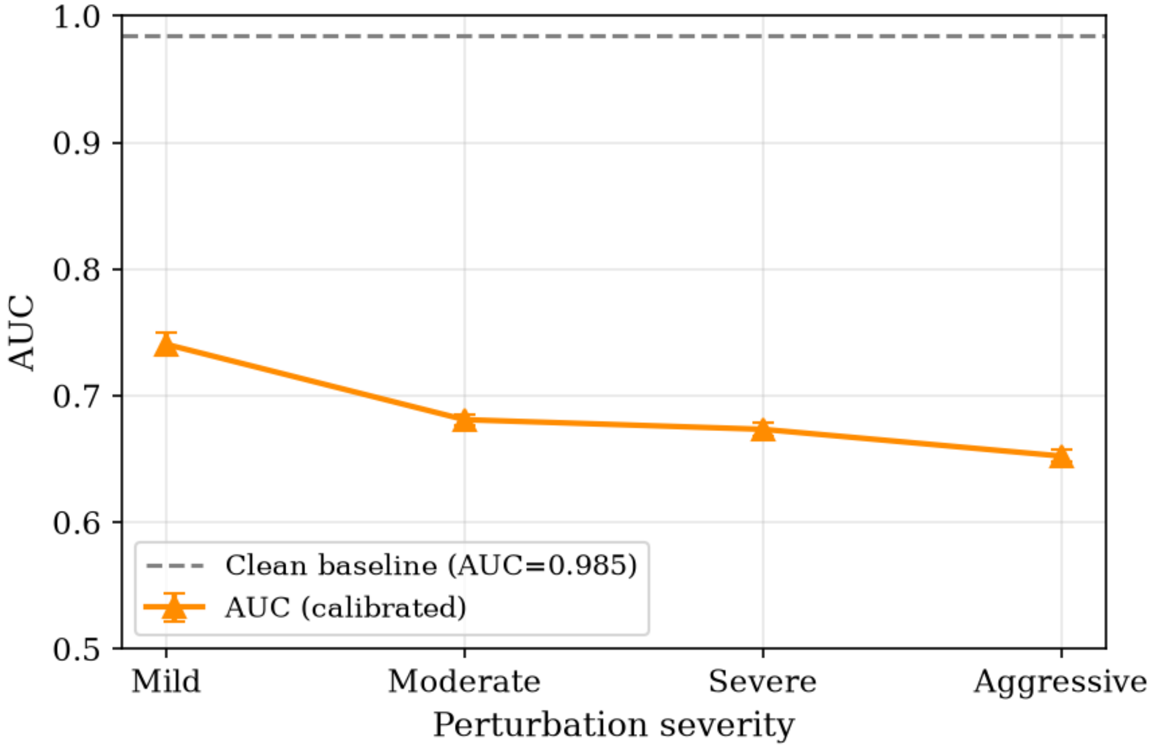


Figure 3: AUC across severity levels. Discrimination degrades more gracefully than calibration, confirming the model retains rank-ordering ability.

Table 6: Results at $H = 30$.

Condition	ECE (raw)	ECE (cal)	ECE (recal)	AUC (cal)
Clean	—	0.013	—	0.994
S1	0.233±0.004	0.288±0.009	0.083±0.021	0.752±0.003
S2	0.235±0.010	0.298±0.011	0.070±0.017	0.706±0.004
S3	0.227±0.011	0.294±0.011	0.068±0.023	0.688±0.001
S4	0.265±0.010	0.317±0.005	0.070±0.027	0.678±0.008

Table 7: Results at $H = 50$.

Condition	ECE (raw)	ECE (cal)	ECE (recal)	AUC (cal)
Clean	—	0.023	—	0.998
S1	0.320±0.016	0.331±0.011	0.085±0.025	0.757±0.005
S2	0.343±0.018	0.366±0.013	0.076±0.022	0.742±0.002
S3	0.343±0.018	0.370±0.013	0.072±0.024	0.722±0.003
S4	0.359±0.017	0.381±0.012	0.079±0.034	0.731±0.005

4.4 Recalibration Recovery

Recalibrating isotonic regression on a 10% sample of perturbed data recovers calibration across all conditions (Table 8). ECE drops to 0.06–0.09—a 71–88% reduction (evaluated on held-out data). The recovery is consistent regardless of severity or horizon.

Table 8: Recalibration recovery across all horizons (held-out evaluation).

Horizon	Clean ECE	Perturbed range	Recalibrated range
10	0.010	0.272–0.313	0.057–0.076
20	0.031	0.277–0.293	0.060–0.073
30	0.013	0.288–0.316	0.076–0.092
50	0.023	0.331–0.381	0.080–0.094

4.5 Root Cause: Feature Shift

The primary driver of calibration degradation is the shift in minimum voltage. In full discharge, $V_{\min}$ reaches approximately 2.3 V. At 75% DoD truncation (Severity 1), it rises to approximately 3.2 V—a 39% increase. The XGBoost model learned split thresholds on clean feature ranges, and the isotonic calibrator cannot adapt to the shifted raw probability distribution.

5 Discussion

5.1 Operational Boundary Map

Our results define three operating zones:

- **Safe (ECE $<$ 0.05):** Clean laboratory conditions only.
- **Warning (ECE 0.05–0.10):** Recalibrated operation. Achievable with a 10% operational sample.
- **Unsafe (ECE $>$ 0.10):** Direct deployment under any partial cycling. ECE exceeds 0.27 regardless of severity.

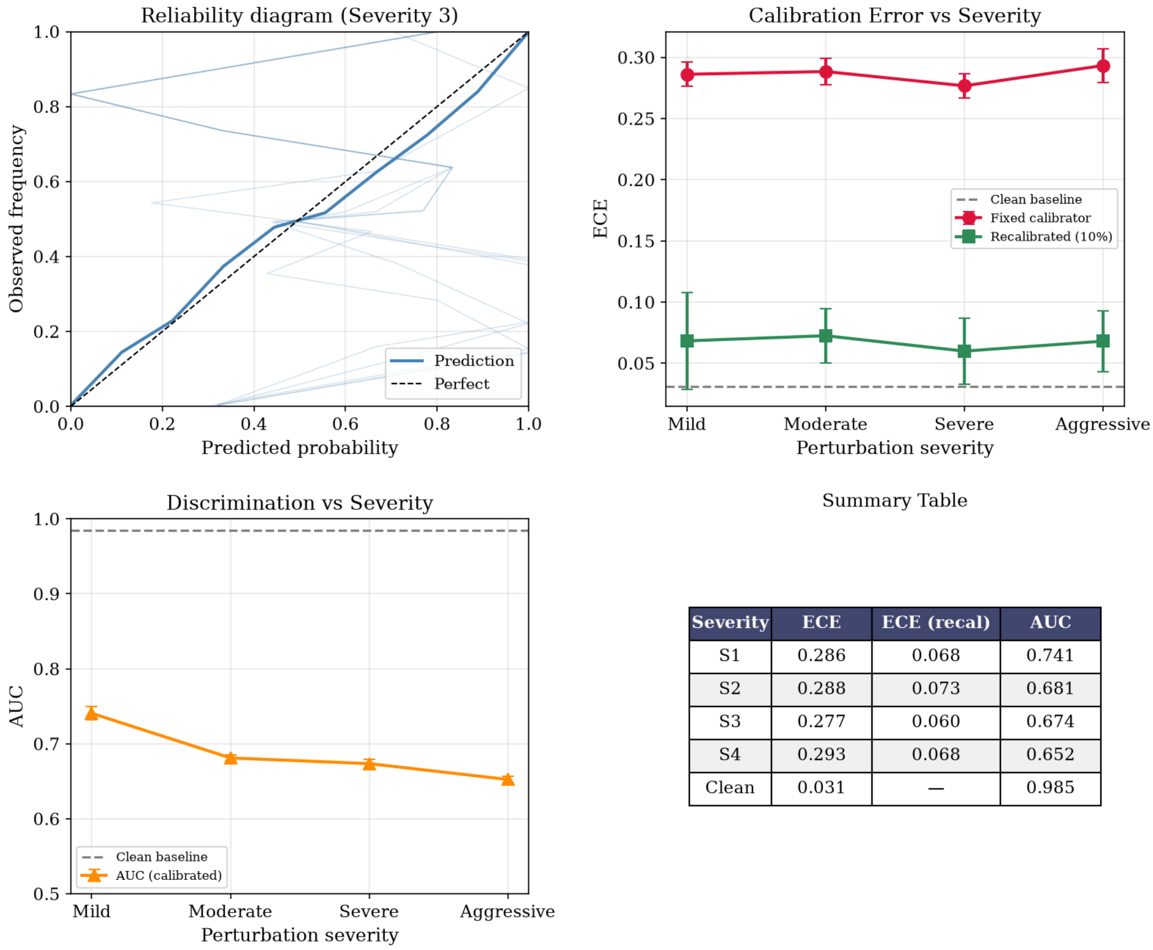
5.2 Why Calibration Is Fragile

The isotonic regression layer, essential for clean-data performance, becomes the weakest link under distribution shift. The saturation effect means even minimal partial cycling causes near-maximal degradation. AUC degrades less severely (0.985 to 0.65–0.74), meaning the model retains rank-ordering ability—it is the probability magnitude that becomes unreliable.

5.3 Practical Recalibration Strategy

A field operator deploying the published model would:

1. Collect 50–100 operational cycles from normal BESS operation.
2. Extract the same seven per-cycle features.



Severity	ECE	ECE (recal)	AUC
S1	0.286	0.068	0.741
S2	0.288	0.073	0.681
S3	0.277	0.060	0.674
S4	0.293	0.068	0.652
Clean	0.031	—	0.985

Figure 4: Combined robustness summary. The left panel shows raw ECE, calibrated ECE, and recalibrated ECE across all conditions; the right panel shows the corresponding AUC values. Recalibration consistently recovers calibration across all horizons and severity levels.

3. Run the frozen XGBoost model to get raw probabilities.
4. Fit a new isotonic regression on raw probabilities vs. observed outcomes.
5. Deploy the recalibrated model.
6. Periodically update the calibrator.

This requires no base classifier retraining, no original training data, and no additional feature engineering.

5.4 Limitations

Five limitations should be noted. First, we preserve the true SOH trajectory, isolating observational shift but not capturing coupled degradation dynamics. Second, results are for LCO chemistry only; validation across LFP, NMC, and other chemistries is needed. Third, we test only 10% recalibration; optimal sample size needs characterization. Fourth, the Gaussian temperature noise model does not capture systematic thermal gradients or diurnal variation. Fifth, only XGBoost is tested; other architectures may differ.

6 Conclusion

This paper presents the first systematic robustness evaluation of multihorizon battery failure intelligence under realistic operating conditions. Three principal findings emerge:

1. Calibration degrades substantially under any level of partial cycling. ECE increases from 0.01–0.03 to 0.27–0.38, saturating at the mildest perturbation.
2. Discrimination degrades moderately (AUC 0.99–1.00 to 0.63–0.76). The feature extractor generalizes better than the calibrator.
3. Recalibration on a 10% operational sample (held-out evaluation) recovers 71–88% of the calibration gap (ECE 0.06–0.09), providing a practical path to field deployment.

Multihorizon hazard models are not inherently robust to observational distribution shift, but a lightweight recalibration strategy suffices to maintain reliability in the field.

References

[1] T. A. Shikdar and H. Laaksonen, "Learning when not to use a battery: Multihorizon failure intelligence," *International Transactions on Electrical Energy Systems*, vol. 2026, p. 6000810, 2026.

[2] B. Saha and K. Goebel, "Battery data set," NASA Ames Prognostics Data Repository, 2007.